

AoC_06 — On Gravity (α^G , G) [STUB]

Appx_07_AoC_06_On_Gravity_PF_v0.1

Status: Stub (structural only)

Role: Atlas build-order continuation; survivability target declaration

Non-claim: No completed derivation yet

Goal:

Define gravity as a survivable coherence-dilution invariant, not a fundamental force or tuned parameter.

Establish the derivation program, dependency spine, and falsification hooks required to derive the dimensionless gravitational coupling α^G , with G recovered secondarily as a unitful mapping constant.

Outputs (Planned)

- Formal definition of α^G as a long-mode coherence residual
- Explicit separation of:
 - dimensionless survivor (α^G)
 - unitful conversion constant (G)
- Dependency map linking gravity to:
 - mass survivors (proton / neutron)
 - collapse operator
 - dimensional ordering
- Survivability window constraints (overbinding vs underbinding)
- Falsification pathways (scale, epoch, strong-curvature regimes)

Dependencies

Required prior AoC entries:

- AoC_00 — The Collapse Operator
- AoC_01 — π
- AoC_02 — c
- AoC_03 — α (1/137)
- AoC_04 — Proton Mass
- AoC_05 — Neutron Mass

Required appendices:

- Appx_10.a — Gravity Threshold

- Appx_10 — HoQG
- Appx_13 — GR Reduction
- Appx_08 / 09 — Time & Spatial Dimensionality
- Appx_16 — CCT-1
- Appx_17 — Linda Function
- Appx_18 — Collapse Operator (formal)
- Appx_20 / 21 — Black Holes & Experiments

Collapse Consequence

If gravity cannot be derived as a selected coherence-dilution invariant, then its weakness must be imposed by hand.

At that point:

- force hierarchy becomes arbitrary,
- fine-tuning reappears,
- long-lived structure loses explanation,
- and the Atlas of Constants fails to enforce consequence.

Notes

- This entry is intentionally skeletal.
- It exists to hold place in the AoC build order.
- Promotion from stub requires:
 1. explicit coherence measure,
 2. long-mode suppression law,
 3. derivation of α^G without importing G,
 4. survivability window analysis,
 5. falsifiers tied to observation.